
Francesco Zamponi

Associate Professor of Theoretical Physics

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francescozamponi.github.io

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Positions

09/2023–present	Professore Associato, Dipartimento di Fisica, Sapienza Università di Roma
10/2018–08/2023	Directeur de Recherche (DR2), CNRS, LPENS, École Normale Supérieure, Paris
10/2018–08/2021	Professeur associé, Physics Department, École Normale Supérieure, Paris
09/2015–09/2018	Maître de Conférences associé, Physics Department, École Normale Supérieure, Paris
03/2012–09/2018	Chargé de Recherche (CR1), CNRS, LPT, École Normale Supérieure, Paris
03/2008–02/2012	Chargé de Recherche (CR2), CNRS, LPT, École Normale Supérieure, Paris
10/2007–02/2008	EU Marie Curie Fellowship, LPT, École Normale Supérieure, Paris
10/2006–10/2007	Post-doc, SPhT Saclay, Paris
10/2005–10/2006	Post-doc, LPT, École Normale Supérieure, Paris
11/2002–10/2005	PhD student, Physics Department, Sapienza Università di Roma Supervisors: Prof. Giancarlo Ruocco and Prof. Giorgio Parisi
01/2002–10/2002	INFM fellowship, Sapienza Università di Roma

Degrees and qualifications

2023	Abilitazione Scientifica Nazionale (fascia I), sector 02/D1 (valid until 11/01/2035)
2013	Abilitazione Scientifica Nazionale (fascia I), sector 02/A2 (valid until 14/04/2033)
2013	Abilitazione Scientifica Nazionale (fascia I), sector 02/B2 (valid until 11/05/2033)
2012	Habilitation à diriger des recherches, École Normale Supérieure, Paris
2006	PhD in Physics, Sapienza Università di Roma
2001	Laurea in Fisica, cum laude, Sapienza Università di Roma

Awards, grants and long-term visiting positions

2024	Sapienza Award Horizon Europe, for the MSCA Doctoral Network proposal <i>Fitness landscapes from molecules to life</i> (FITMOLIFE)
2024	Sapienza Progetto di Ricerca Medio, <i>Stochastic dynamics of biological systems</i>
2016	ERC Consolidator Grant, <i>GlassUniversality</i>
2016	PI of the Simons Collaboration <i>Cracking the Glass Problem</i>
2014	PSL* University grant, <i>Jamming: Theory and Experiment</i> , with O. Dauchot
2013	Appointment as regular visiting fellow, Duke University, Durham, NC
2011	PIR grant of ENS, <i>Optimization in complex systems</i>
2010	MIT-France Seed Fund / MISTI Global Seed Fund grant, <i>Numerical simulations and quantum algorithms</i>
2009	Visiting fellow, Princeton Center for Theoretical Science, Princeton, NJ (09–12/2009)
2009	BQR grant of ENS, <i>Fluids and solids at low temperature</i>
2007	“Giorgio Gamberini” award of Scuola Normale Superiore, Pisa, for a PhD thesis in theoretical physics discussed after January 2005
2002	INFM–Sapienza prize for an experimental Laurea thesis discussed in 2001

Editorial work, committees and scientific service

2025–present	Member of the editorial board, <i>Journal of Statistical Mechanics: Theory and Experiment</i> , IOP
2025	Member of the PE3 Consolidator Grant panel, European Research Council
2021–present	Associate editor, <i>Journal of Statistical Physics</i> , Springer
2019–2021	Elected head of the Statistical Physics group, LPENS, Paris
2015–2016	Elected member of the Commission interdisciplinaire 51, CNRS
2014	Member of the selection committee for a Maître de Conférences position, LPT-Orsay
2013	Member of the AERES visiting committee, LPT-Orsay
2012–2016	Elected member of the Comité national, Section 02, CNRS
2010–2014	Member of the scientific committee of the GDR Phenix, CNRS

Refereeing

Nature, Nature Physics, Nature Machine Intelligence, Nature Materials, Nature Communications, PNAS, eLife, PLoS Computational Biology, Current Opinion in Genetics & Development, Physical Review Letters, Physical Review E, Journal of Statistical Physics, Journal of Statistical Mechanics: Theory and Experiment, Journal of Chemical Physics, Journal of Physics: Condensed Matter, European Physical Journal B, Europhysics Letters, Soft Matter, Molecular Physics, Physica A, Journal of Alloys and Compounds, Powder Technology. See [ORCID](#) for additional information.

Research

Research interests	francescozamponi.github.io/pagine/research.html
Publications	francescozamponi.github.io/pagine/pubblicazioni.html
Talks	francescozamponi.github.io/pagine/talks.html
Software	francescozamponi.github.io/pagine/software.html
Events organized	francescozamponi.github.io/pagine/events.html

Long-term research visits

Princeton University 2009 (107 days); Sapienza Rome 2011–2014 (93 days over five visits); SISSA Trieste 2008–2010 (52 days over four visits); Porto Alegre University 2010, 2011 (22 days); MIT Boston 2010, 2011 (19 days); Duke University 2013 (8 days); Weizmann Institute 2013 (5 days); Graduate Center, CUNY 2014 (3 days).

Teaching

2026–present	<i>Statistical inference and biological data analysis</i> , 60h, master, Physics, Sapienza
2023–present	<i>Meccanica analitica e relativistica</i> , 60h, bachelor, Physics, Sapienza
2023–2026	<i>Modelli matematici per la fisica II</i> , 56h, bachelor, Mathematics, Sapienza
2020–2023	<i>Statistical Physics 2: Disordered Systems and Interdisciplinary Applications</i> , 20h, with G. Schehr, ICFP master (M2), ENS Paris
2016–2021	<i>Information, inference, networks: from statistical physics to big biological data</i> , 20h, with R. Monasson and S. Cocco, ICFP master (M2), ENS Paris
2015–2019	<i>Mathématiques pour physiciens</i> , 38h, with M. Petrini, L3, ENS Paris
2017	<i>Machine learning</i> , 6h, with F. Krzakala and A. Manoel, CFM, Paris
2011–2016	<i>Introduction à la physique numérique</i> , 12h, with V. Croquette, L3, ENS Paris
2008–2012	<i>Mécanique statistique</i> (préceptorat), 16h, assistant to J.-F. Joanny, L3, ESPCI Paris
2007–2014	<i>Spin glasses</i> , PhD programmes of SISSA Trieste (20h, 2008–2010) and Sapienza Rome (20h, 2011–2014). Lecture notes: arXiv:1008.4844
2004–2005 and 2006–2007	<i>Fisica dei liquidi</i> , 5h, assistant to P. Tartaglia, master, Sapienza
2003–2004	<i>Metodi matematici della fisica</i> , 40h, assistant to A. Degasperis, bachelor, Sapienza

Supervision

PhD students

Years	Student	Institution	Co-supervisor	Current position
2026–29	Federico Celauro	Sapienza Rome	F. Ricci-Tersenghi	
2025–28	Ottavia Schulte	Sapienza Rome	R. Di Leonardo	
2023–26	Lorenzo Rosset	ENS Paris	M. Weigt	
2023–26	Alya Zeinaty	Sorbonne Université	M. Weigt	
2023–26	Luca Del Bono	Sapienza Rome	F. Ricci-Tersenghi	
2022–26	Riccardo Cipolloni	Sapienza Rome	F. Ricci-Tersenghi	
2020–23	Jeanne Trinquier	Sorbonne Université	M. Weigt	Postdoc, Institut de la Vision, Paris
2020–23	Matteo Bisardi	ENS Paris	M. Weigt	Postdoc, Univ. of British Columbia
2020–23	Enrico Ventura	Sapienza Rome	G. Ruocco	Postdoc, SISSA Trieste
2017–20	Dhruv Sharma	ENS Paris	J.-P. Bouchaud	Talagent Financial, Boulder, CO
2016–19	Camille Scalliet	Univ. Montpellier	L. Berthier	Permanent, CNRS – LPENS Paris
2013–16	Thibaud Maimbourg	ENS Paris	J. Kurchan	
2012–15	Corrado Rainone	Sapienza Rome	G. Parisi	Qualcomm, Amsterdam
2011–12	Laetitia Papaxanthos	ENS Paris	J.-P. Bouchaud	(PhD at ENS not completed)
2010–13	Victor Bapst	ENS Paris	G. Semerjian	Google DeepMind, London
2009–11	Laura Foini	SISSA Trieste	A. Gambassi	Permanent, CNRS – IPhT Paris

Post-doctoral researchers

Years	Researcher	Funding	Current position
2023–26	Saverio Rossi	Overheads	Postdoc, Cologne
2021–22	Sabrina Cotogno	Simons	Researcher, Capgemini Engineering
2020–23	Simone Ciarella	Simons, ERC	Permanent, Netherlands eScience Center
2020–23	Felix Mocanu	Simons, ERC	Postdoc, University of Oxford
2020–22	Giampaolo Folena	Simons, ERC	Postdoc, Spain
2020	Misaki Ozawa	ERC	Permanent, CNRS – LiPhy Grenoble
2019	Davide Facoetti	ERC	Research scientist, Faculty AI
2018–21	Alessandro Manacorda	ERC	High school teacher, Rome
2018–19	Anna Paola Muntoni	Simons	Associate Professor, Politecnico di Torino
2017–20	Harukuni Ikeda	Japanese grant, Simons	Associate Professor, Tokyo University
2017–19	Dmytro Khomenko	Simons	Postdoc, Poland
2017–18	Ada Altieri	ERC	Maître de Conférences, Université de Paris
2016–18	Elisabeth Agoritsas	Simons, ERC	High school teacher, Geneva
2015–18	Beatriz Seoane Bartolomé	Marie Curie fellowship, ERC	Professor, UCM Madrid
2012–14	Yuliang Jin	PSL grant	Permanent, ITP-CAS, Shanghai
2012–14	Stanislao Gualdi	Crisis Project	Capital Fund Management, Paris

Master, Laurea and internship students

Year	Student	Stage	Institution	Co-supervisor
2026	Federico Celauro	M2	Sapienza Rome	F. Ricci-Tersenghi
2026	Raphaël Decan de Chatouville	M1	ENS Paris	—
2026	Iker Landa-Cunha	M1	ENS Paris-Saclay	—
2024	Guglielmo Mennella	M2	Sapienza Rome	P. Charbonneau
2024	Gabriele Farné	M2	Sapienza Rome	—
2024	Shoichi Yip	M2	Sapienza Rome	C. Nizak, M. Weigt
2024	Ottavia Schulte	M2	Sapienza Rome	—
2023	Leonardo Di Bari	M2	Sorbonne Université	M. Weigt
2020	Jeanne Trinquier	M2	Université de Paris	M. Weigt
2020	Matteo Bisardi	M2	Université de Paris	M. Weigt
2017	Dhruv Sharma	M2	ENS Paris	J.-P. Bouchaud
2017	Guillaume Nevo	M2	ENS Paris	J.-P. Bouchaud
2017	Steffen Arnold	Internship	ENS Paris	—
2015	Matthieu Mangeat	M2	ENS Paris	—
2015	Giancarlo Croce	M1	ENS Paris	S. Cocco
2015	Alexis Poncet	M1	ENS Paris	E. Corwin
2014	Giulia Cencetti	M2	ENS Paris	—
2014	Marcus Cordi	M2	KTH Stockholm	J.-P. Bouchaud
2013	Giulia Carra	M2	ENS Paris	P. Charbonneau
2011	Nicolas Allegra	M2	Master “Systèmes Complexes”	A. Walczak
2011	Vishal Agarwal	Internship	IIT Guwahati, India	—
2009	Indaco Biazzo	Laurea	Sapienza Rome	G. Parisi
2009	Hugo Jacquin	M2	Master “Systèmes Complexes”	—
2008	Francesco Caltagirone	Laurea	Sapienza Rome	G. Parisi
2004	Alessio Andronico	Laurea	Sapienza Rome	G. Ruocco

Education: courses attended

Courses followed during the undergraduate and PhD studies at the Department of Physics, Sapienza Università di Roma.

1997–2000	Standard courses of the first three years of the undergraduate course in Physics
2000–2001	<i>Meccanica statistica</i> (Statistical Mechanics), Prof. C. Di Castro, 120 h <i>Fisica dei sistemi disordinati</i> (Physics of Disordered Systems), Prof. E. Marinari, 120 h <i>Fisica teorica III</i> (Quantum Electrodynamics), Prof. N. Cabibbo, 120 h <i>Fisica dei solidi</i> (Solid State Physics), Prof. L. Pietronero, 120 h <i>Fisica delle basse temperature</i> (Quantum Many-Body Systems), Prof. C. Castellani, 120 h <i>Programmazione orientata agli oggetti</i> (Object-oriented Programming), Prof. G. Organtini, 30 h
2001–2002	<i>Metodi matematici della fisica, corso avanzato</i> (Stochastic Processes), Prof. G. Jona-Lasinio, 120 h <i>Meccanica superiore</i> (Statistical Mechanics, Dynamical Systems), Prof. G. Gallavotti, 120 h
2002–2003	<i>Introduzione al modello standard e alle sue estensioni supersimmetriche</i> , Dr. E. Franco and Dr. L. Silvestrini, 60 h <i>Fisica dei sistemi disordinati</i> (Physics of Disordered Systems, for PhD students), Dr. A. Cavagna and Dr. F. Ricci-Tersenghi, 60 h <i>Teoria dei campi</i> (Field Theory), Prof. M. Testa, 80 h

Schools and conferences attended

- 8th International Workshop on Disordered Systems, Andalo (Trento), Italy, 12–15 March 2001.
- Unifying Concepts in Glass Physics, conference, Accademia dei Lincei, Rome, Italy, 26 February–2 March 2002.
- Field Theory and Statistical Mechanics, conference, Department of Physics, Sapienza Rome, Italy, 10–15 June 2002.
- Spectroscopic Investigation of the Collective Dynamics in Disordered Systems, summer school, ICTP Trieste, Italy, 17–28 June 2002.
- Slow Relaxation and Nonequilibrium Dynamics in Condensed Matter, summer school, École de Physique des Houches, France, 1–26 July 2002.
- Perspectives in Mathematical Physics, conference, Department of Physics, Sapienza Rome, Italy, 4–7 September 2002.
- 9th International Workshop on Disordered Systems, Molveno (Trento), Italy, 10–13 March 2003.
- The Physics of Complex Systems (New Advances and Perspectives), summer school, International School of Physics “Enrico Fermi”, Course CLV, Varenna, Italy, 1–11 July 2003.
- Supersymmetry and Matrix Models, XII National Seminar in Theoretical Physics, INFN, Parma, Italy, 1–12 September 2003.
- Applications of Random Matrices to Physics, Marie Curie Training Course, École de Physique des Houches, France, 6–25 June 2004.
- First Latin-American School and Conference on Statistical Mechanics and Interdisciplinary Applications, Havana, Cuba, 28 February–12 March 2005.
- Recent Progress in Glassy Physics, workshop, Institut Henri Poincaré, Paris, France, 27–30 September 2005.
- Jamming, workshop, Aspen Center for Physics, Aspen (CO), USA, 29 July–13 August 2007.